

IoT fueled AI system for Food Wastage Reduction and Efficient Redistribution of Excess Food

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Abstract—Real-time data monitoring aided with Internet of Things (IoT) and analysis of the data fueled by Artificial Intelligence (AI) will play a key role in predicting food consumption pattern and effective redistribution of excess food. This paper proposes a centralized real-time IoT-enriched system, supplemented by past data to predict the real-time food supply and demand in a region. Data will be obtained from the digitized food directory. The data obtained would be, the type of food ordered, the type of day it was ordered, other influential factors and the edge computed data from the sensors which would give the people count. These past data will be analyzed and fed to a model that can predict quantity of food that will be needed every day at a particular place. Data can also be used to estimate the average amount of food that can go as waste every day and come up with adequate logistics to transfer this surplus food to places in need of food without it going stale.

Index Terms—AI, IoT, Data processing, Food distribution, ML, Deep Learning, Food supply chain, Food Wastage.

I. INTRODUCTION

According to the Food and Agriculture Organization (FAO) report entitled 'The State of Food Security and Nutrition in the World 2020' almost 690 million people worldwide (8.9% of the world's population) are estimated to have been undernourished in 2019 [1]. That said, the global volume of edible food waste is estimated at 1.3 billion tonnes [2]. This waste is not food waste alone, the equivalent carbon footprint is 3.3 billion tonnes, the water needed to generate this amount of food is 250 cubic km, that is 3 times the volume of Lake Geneva and about 28% of the total agricultural land available is used to cultivate this waste food. There are five stages in the food supply chain (FSC), they are agricultural harvest, post-harvest handling and storage, production processing and packaging, distribution and consumption. Food is wasted in all of these stages in different ways. Worldwide, food waste impacts sustainability and efficiency of food and nutrition programs.

On one hand, the surplus food is getting wasted while on the other hand millions of people are hungry. How can technology help in this context? How can technology be applied to overcome this situation? Can a system be devised to distribute this surplus food which can be used to feed the needy? This system would help in reducing the food wasted, thus contributing to the sustainable development of the world

and at the same time, would feed the hungry. This paper suggests such a novel system aided with technology. Existing ideas that use technology for reducing the food wastage are first highlighted and then an efficient model is suggested which if implemented on a large scale would help to effectively reduce the food wastage and will aid in redistribution of the excess food to the needy.

II. TECHNOLOGY IN FOOD WASTE REDUCTION

Over the years, innovations such as Machine Learning (ML), AI and Data Science have seen quantum strides that have developed into something amazing. They have been deployed to help people make the right decisions. These intelligent systems have marked their presence in a variety of other industries around the world and their implementations in all fields are growing rapidly. Food industry is one of the new markets where AI and ML are being used. There are many food products in the supermarkets that go to waste every day, mainly because of weak demand forecasting. ML will play a crucial role in this scenario and reduce the amount of food waste. ML will create a better demand forecast using previous data on how much food goes to waste, how much a product sells etc. It will allow supermarket owners to order products at an acceptable level, not more or not less. Intelligent and creative processes will certainly enhance demand preparedness in the supply chain. IoT technology is trying to deliver solutions that enhance the supply chain by using a wide range of sensors, some with very high precision measurements and ML algorithms for Big Data analysis. One of the companies operating in this direction is Centaur Analytics, which began in Volos, Greece in 2014. They aim to provide a system that will upgrade the quality of the post-harvest stage of the supply chain [3].

There are a few innovative AI restaurant analytic firms. These companies help a restaurant to manage the raw food material to be purchased on a weekly basis and also predicts the consumer inflow pattern. Thus, efficiently and technically managing the food resource. One such company is Tenzo [4]. They predict the future food demand with the power of ML and AI to order the right amount of food needed by taking into account data from a range of factors - including account sale, weather, social media reviews etc. Weighing in these factors,

the software predicts the footfall in a particular place at a particular time and eliminate the food waste. Real-time data and machine learning are providing appealing solutions to the restaurants. One can see a fast paced adoption of AI and ML technologies in the food industries.

AI and IoT solutions are already being deployed in the Food Industry with positive outcomes. These proven technologies could be combined so that food wastage and world hunger could be tackled.

III. PROPOSED MODEL

This paper proposes an IoT fueled AI system for food wastage reduction and efficient redistribution of excess food. This system is divided into 4 stages, Stage 1 consists of the places where the excess food would be present like restaurants and marriage halls. These places would be equipped with an array of sensors to collect data that will be transferred to a central server via IoT. The server will also have data from sensors located at places where food is in need (stage 4) such shelter homes, old age homes, orphanages etc. The supply

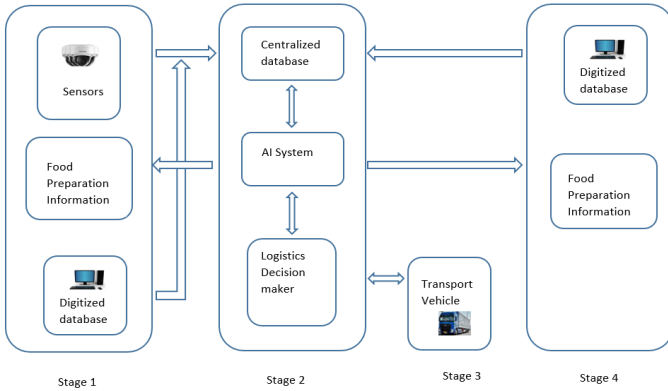


Fig. 1: Constituents of the proposed model

and demand will be dynamic and depends on various factors but they won't be completely random. The AI algorithm in the Stage 3 will come into play here, the system will be trained and modeled initially as per some collected data. This AI system will predict the amount of food that will be left over in a city on an average day and in real-time monitor the amount of food that may go unused and deploy an efficient logistics to distribute the surplus food to the food deficient places. The AI system can further notify these places about the amount of food that may arrive there as donations, they can prepare additional amount of food, if need be. Fig. 1 shows the proposed system's block diagram with all the 4 stages. The functionality and constituents of the four stages are elaborated in the following sections.

A. Stage 1 :- Places with surplus food

Any place where there is a risk of surplus food being available, such as restaurants, wedding/function halls and other social gathering venues. If these locations deploy this model, they have dual advantages.

- One is that, this system based on past records and other variables, will forecast and report footfall to these locations on a given day. This data would help these places to prepare food for only those many people.
- Second is to make use of the extra food, that remains due to some unforeseen reasons. The excess food will be properly collected and redistributed before the food goes stale.

Since the predictions can sway to some extent, a real-time analysis of the footfall and amount of food remaining need to be obtained, once the AI system(Stage 2) has this data the real-time feedback mechanism in the system will notify these places if more food needs to be prepared or not. Stage 1 consists of three sections, they are

1) Sensors:

- CCTV camera data scaled with edge computing. This data gives the people count and would be transmitted to the central server on real-time basis through an IoT device.
- Smart Containers which can determine how much food was prepared and how much food is remaining. This is, again a real-time data and it would be send to the central server. The sufficient volume of food which needs to be prepared is determined and measured.

2) Digitized Database:

- All the details and records of the place, if restaurants then billings, if banquet halls then directory containing future bookings etc. would be in this digitized database. This digitized database would also contain the location details, the day of the week, special locations etc.
- These data are transferred to the IoT device which in turn sends it to server.

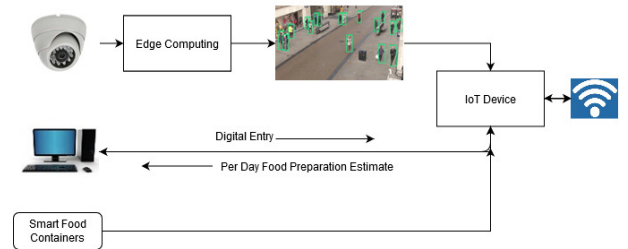


Fig. 2: Constituents of stage 1

3) Food estimated Data :

- This is an input from the second stage of the proposed model. The AI system in the server will predict the amount of food needed for a particular day and conveys it to the place via the IoT device.

Clubbing all the information from all the sections of Stage 1 the AI system running in the server would provide the information about the amount of food which needs to be prepared on a particular day. If the optimum amount of food is prepared, food wastage would be low. The AI system with the real-time input data would be able to predict if any food

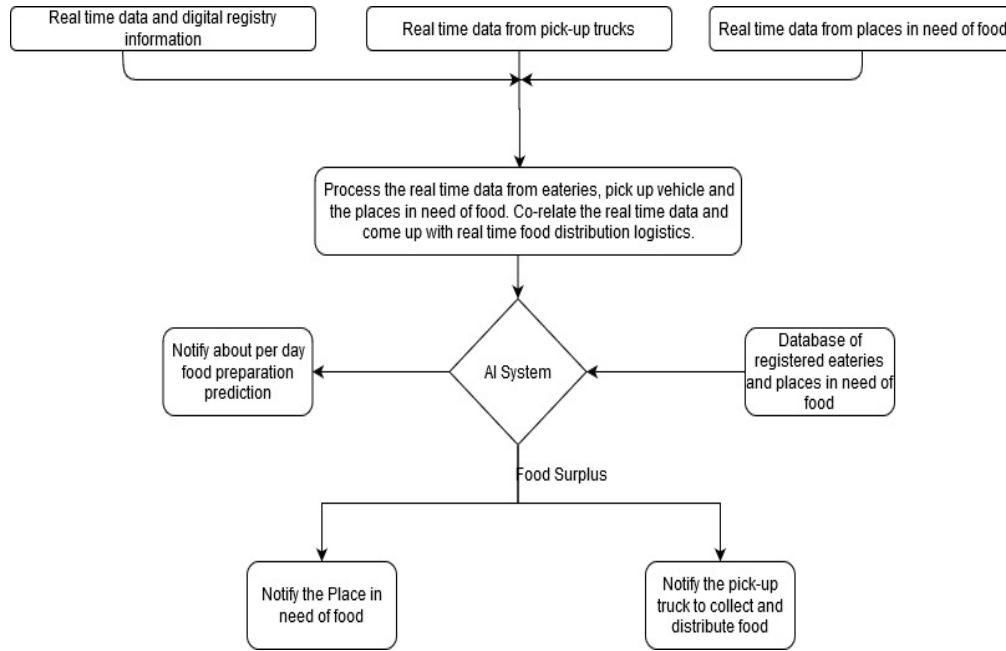


Fig. 3: Overall generic algorithm

can go as waste.

In case of food shortage, the AI system will alert the place in real-time about the need to make more food in order to meet the demand and by what quantity it has to be prepared. It would take into consideration the amount of time that would be needed to prepare the food on time. If the system detects any extra amount of food then the AI would initiate the process to collect that excess food and deliver it to the places in need of food before it goes stale. Fig. 2 shows the constituents of Stage 1 and the data flow.

B. Stage 2 :- The AI system in Server

The central node of this proposed model is the server. This will host the centralized database which would be the brain and memory of this system and the AI algorithms would be running on it. Along with these, the Stage 2 consists of another important section which is termed as Logistic decision maker. Following are the three sections of Stage 2,

1) *Centralized database*: The database will act as the history keeper so that the AI system can learn from it. Following are the features of the centralized database.

- **City-wise database**
Previous data of each eatery, event hall etc. These are transmitted from the Stage 1 and is compressed and stored here. This data is used to train and improve the AI system.
- The data includes, date and day of the week, amount of food needed on a particular day, importance of the day and future bookings.
- Each place will have its own database, also the places where additional food is required will have another database. The places in need of food would send the required data daily.

2) *AI system*: The brain of this model is the AI system. Based on the previous data in the database and the real-time data obtained from various locations, the AI system can determine the food demand and in real-time it would service the food shortage via efficient redistribution of the surplus food. The AI system also informs the Stage 1 about the amount of food that needs to be prepared on a particular day. In other words, it makes the decisions and conveys it to Stage 1 and Stage 3. Also the system should be able to handle the data in the centralized database. One can't have infinite memory space to store all the values, therefore AI must decide which data is relevant and which previous data can be ignored and deleted [5]–[7]. The proposed model AI system would also be very complex and the overall system would be implemented by clubbing various known AI models together.

3) *Logistics Decision Maker*: This section plays a pivotal role in making the logistics for the effective redistribution of the food. Food, being a perishable item, if not transported properly and efficiently would go stale. This section uses the data from the AI system to efficiently lay down the routes that can be taken by the vehicles to reach the destination as quickly as possible.

- AI will be predicting the food usage at a particular place on a particular day, the real-time data from these places will convey the amount of food which will be leftover daily.
- The above data along with input from places which need food is fed to the Logistics Decision Maker.
- Shortest way by which the excess food can be redistributed to places in need is identified.
- Real-time data from the transporting vehicles, this helps in keeping the system updated about the traffic situations

and manages the other resources efficiently on a real-time basis.

The overall algorithm the AI model uses is shown in the fig. 3.

C. Stage 3 :- Transport Vehicles

A smart refrigerated transport vehicle will be a part of this proposed model as shown in fig 4. Following are the features

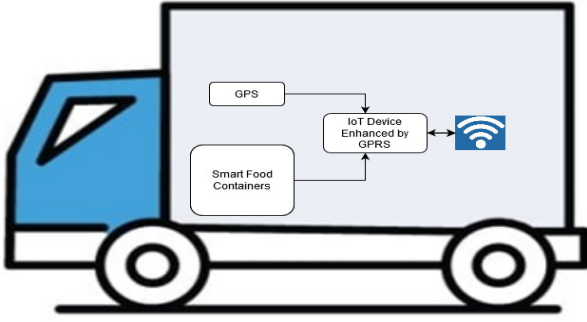


Fig. 4: Smart refrigerated pick-up Truck

of this smart refrigerated transport vehicle:-

- Transmit the real-time location of the vehicle.
The vehicle would be fitted with GPS and GPRS technologies that will transmit its location to the Logistics Decision Maker in Stage 2. This helps in effective routing of the vehicle.
- Real-time food health information is conveyed to the server.
The food would be placed in another type of smart containers, which will send data to the IoT module fitted inside the vehicle, which in turn will transmit the data to the server.
- Optimum route to be taken by the vehicle is obtained from the server's Logistics Decision maker, which sends the information depending upon the real-time location of data being transmitted from this vehicle.

D. Stage 4 :- Places in need of food

The final stage represents the places in need of food like orphanages, old age homes, low income colonies etc. While preparing food at these places the additional food which can come from different places (redistributed food from Stage 1) could be taken into account. The AI system gets data from Stage 4 to determine the approximate number of people who needs to be fed on a particular day in a particular place. The data from Stage 1 would aid the AI system to know what amount of food needs to be redistributed and pass it to Stage 4, resulting in preparation of only that much amount of food, again reducing the wastage of food. Constituents of this stage are :-

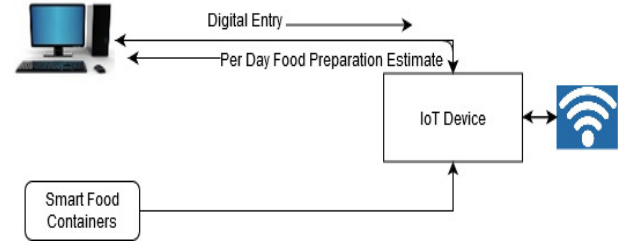


Fig. 5: Constituents of Stage 4

1) *Digitized Database*: Similar to the section in Stage 1 but the data collected would be slightly different. This data gives the information about the number of people who will need food on a particular day. This digital data is send to the server to be saved and the AI model uses this data to train itself.

- All the details and records of the place, like the number of people, who are present, food consumed in a day etc.
- Real-time data like the amount of food left in the smart container and the number of people left to have food is send to the server. The smart container used in this stage is the same as that used in Stage 1. These data are transferred to server on real-time basis.

2) *Food estimated data* : Similar to the section in Stage 1.

- The AI system in the server will predict the amount of food that needs to prepared on a particular day at each location. Remaining food will arrive in transport vehicles. Its timing and quantity will be notified.
- The information would be updated and notified on real-time basis. Based on the real-time data the AI system gets from a location and other factors which can have considerable effect on the amount of food that can reach at a particular place, after processing these and running that through its neural model, the AI system sends back the food estimation data.

Fig. 5 shows the constituents of Stage 4 and the data flow.

The system will become more efficient by continuously learning using the data it gets over time. The ultimate aim of the AI system is not to meet the food requirement at a particular place where it is needed but to reduce the food which is prepared and to effectively redistribute the excess food, if any. This in effect reduces the wastage of food and at the same time feeds the hungry.

IV. CONCLUSION

Approximately 1/3 of the agricultural produce ends up in landfills which is not only the wastage of food but the wastage of energy and other resources like water, agricultural land etc. used to cultivate, transport and prepare the wasted food. A considerable amount of food is wasted after the food reaches the consumer and this excess food can be used to feed the needy. In this paper an innovative system aided with IoT and AI technologies is proposed to tackle the problem. An architectural overview of the proposed system is presented. How

the system could be realized using AI and the type of sensors needed for the realization is also suggested. The proposed self-learning algorithm, on implementation, will gradually reduce the food wastage over time. With the efficient redistribution of excess food, the problem of malnutrition could be solved. When the food wastage reduces, the food prices too would go down which would result in access to healthy and nutritious food for the underprivileged.

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